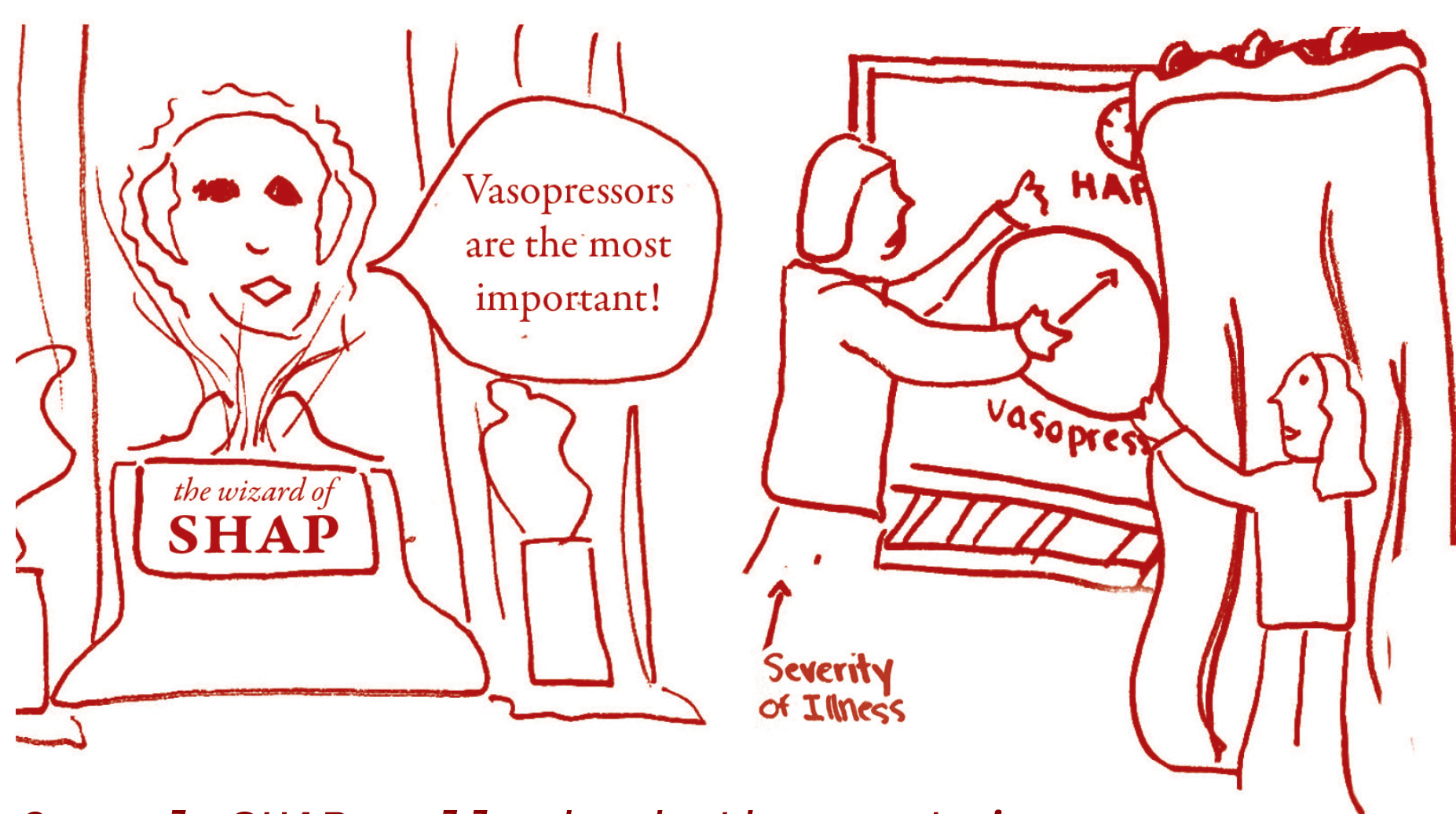




Causal SHAP pulls back the curtain.

THE PROBLEM

SHAP has short memory.

SHAP averages over every feature ordering as if causes and effects were interchangeable. They are not. Mediators arrive after their parents, absorbing residual credit that belongs upstream. The closer to the outcome a feature sits, the more it absorbs. Mediators look important. Confounders look causal. Upstream drivers look weak.

A SOLUTION

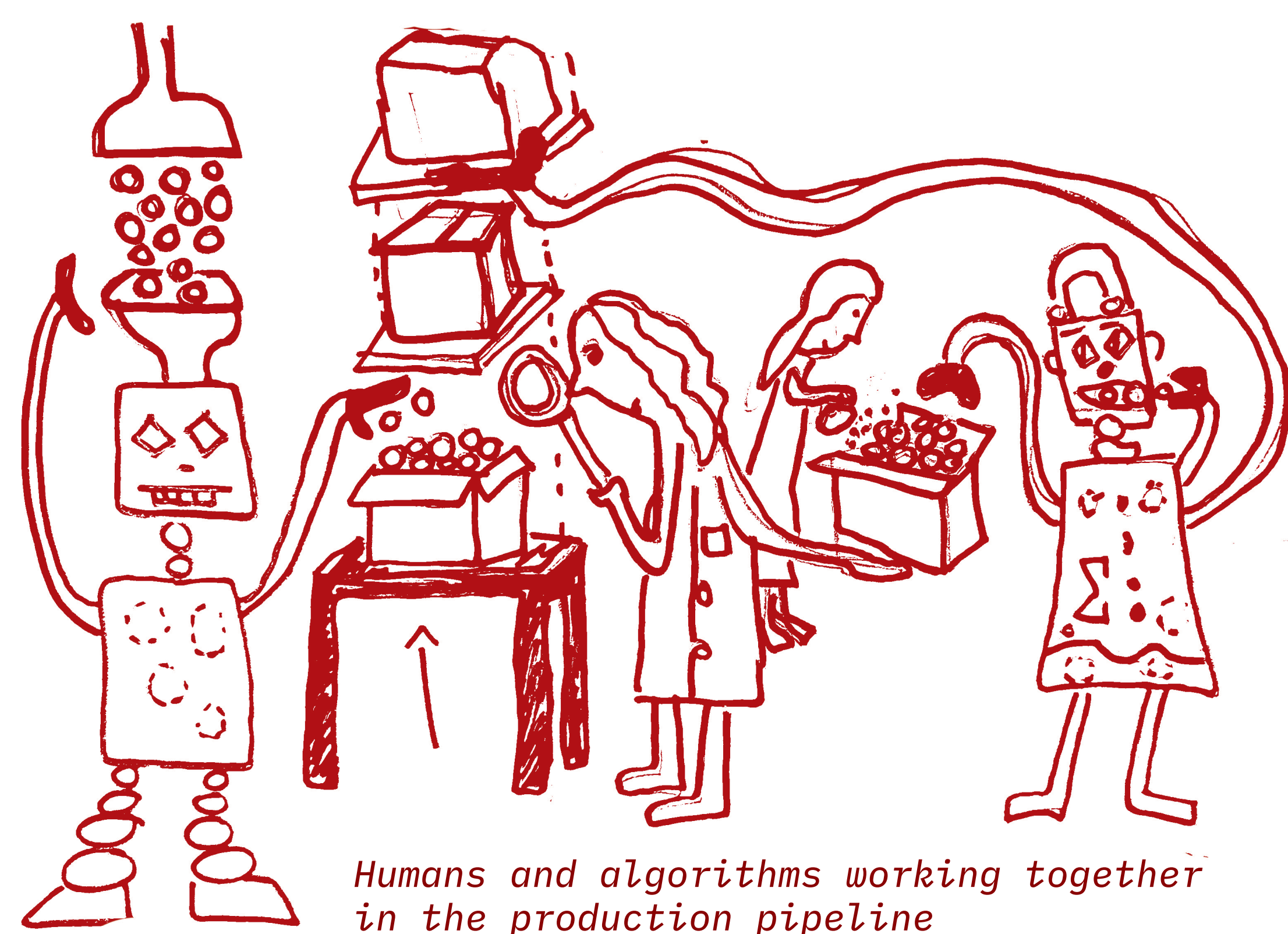
Causal SHAP remembers the DAG.

Heskes (2020) and Janzing (2020) proposed two changes: restrict orderings to those consistent with the DAG, and replace the observational expectation with an interventional one. Credit can no longer flow backward through time or sideways across confounders.

THE WORKFLOW

Humans and algorithms working together.

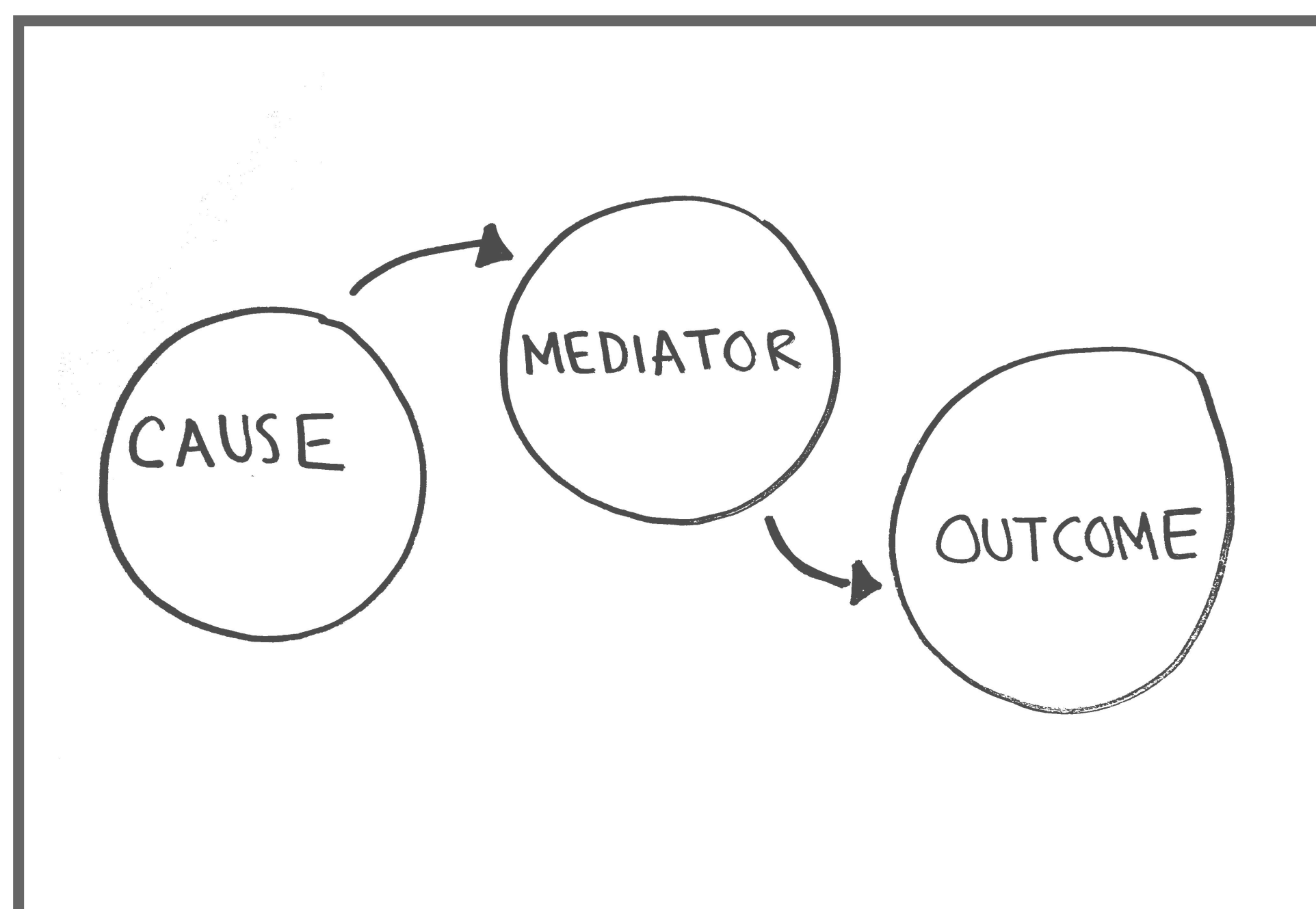
Discovery returns a graph it cannot finish. Causal SHAP needs one that's complete. The expert in the middle orients ambiguous edges. The expert at the end reads the attribution and iterates. The value is the cycle.



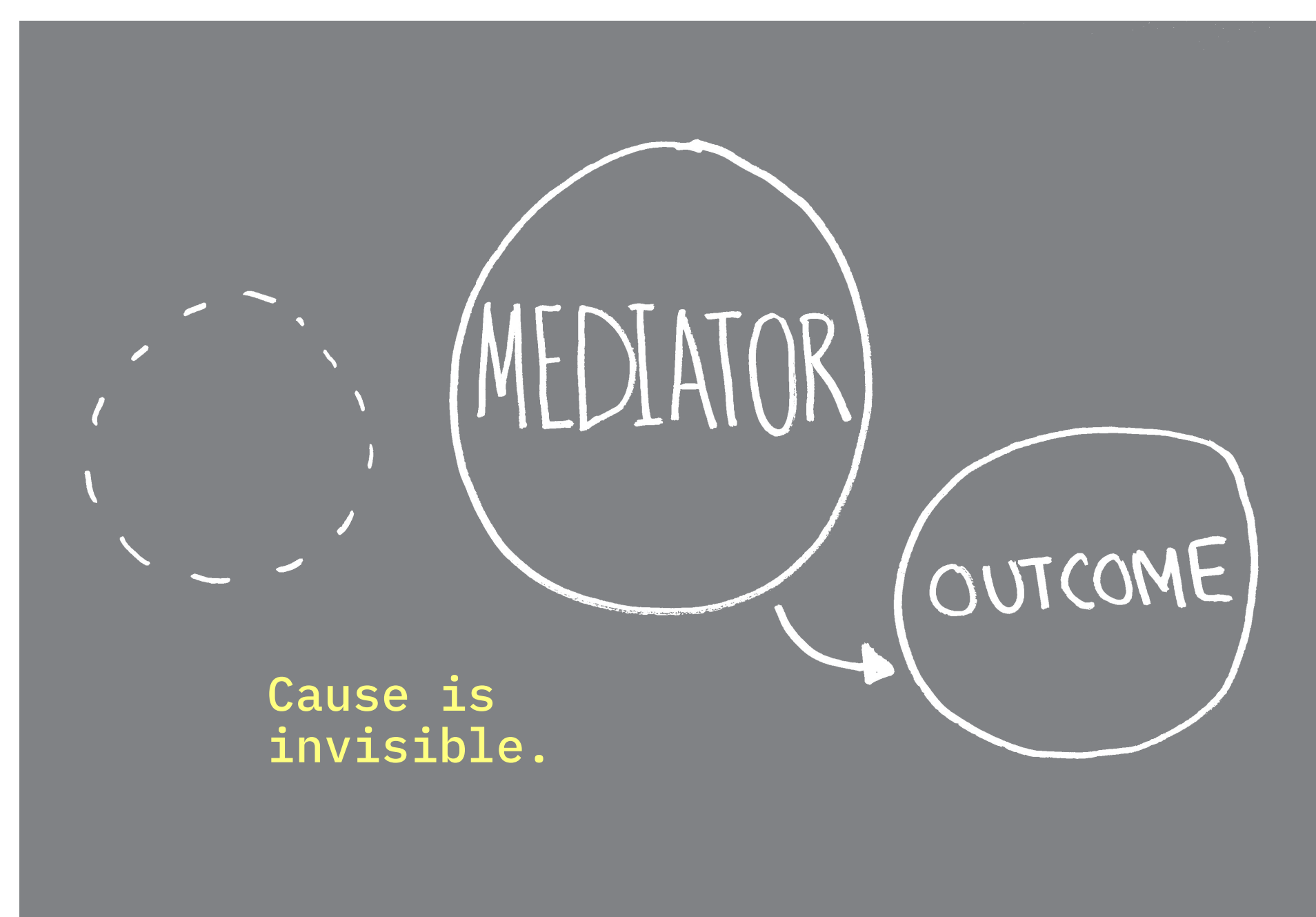
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SHAP has short memory. Causal SHAP remembers the DAG.

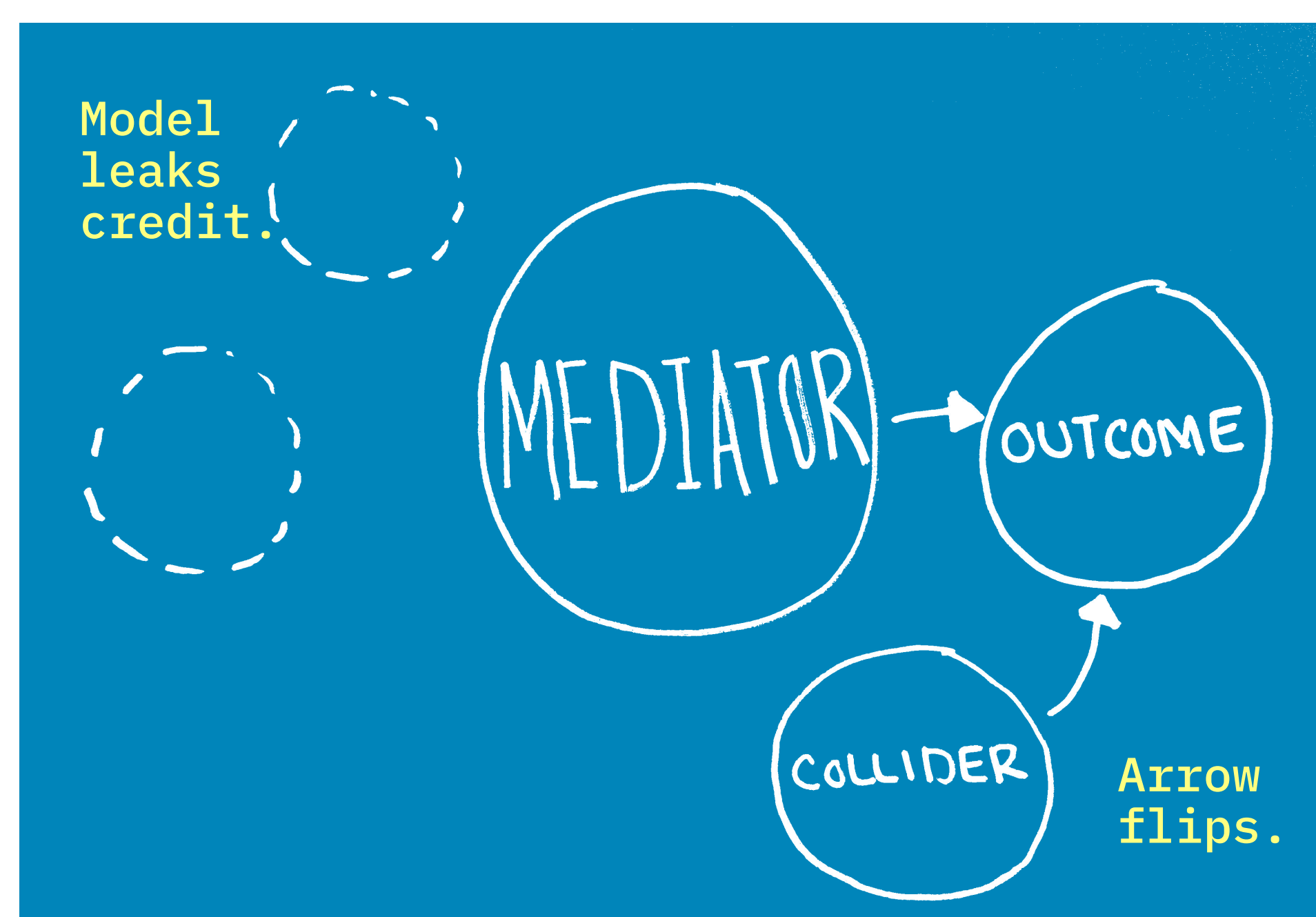
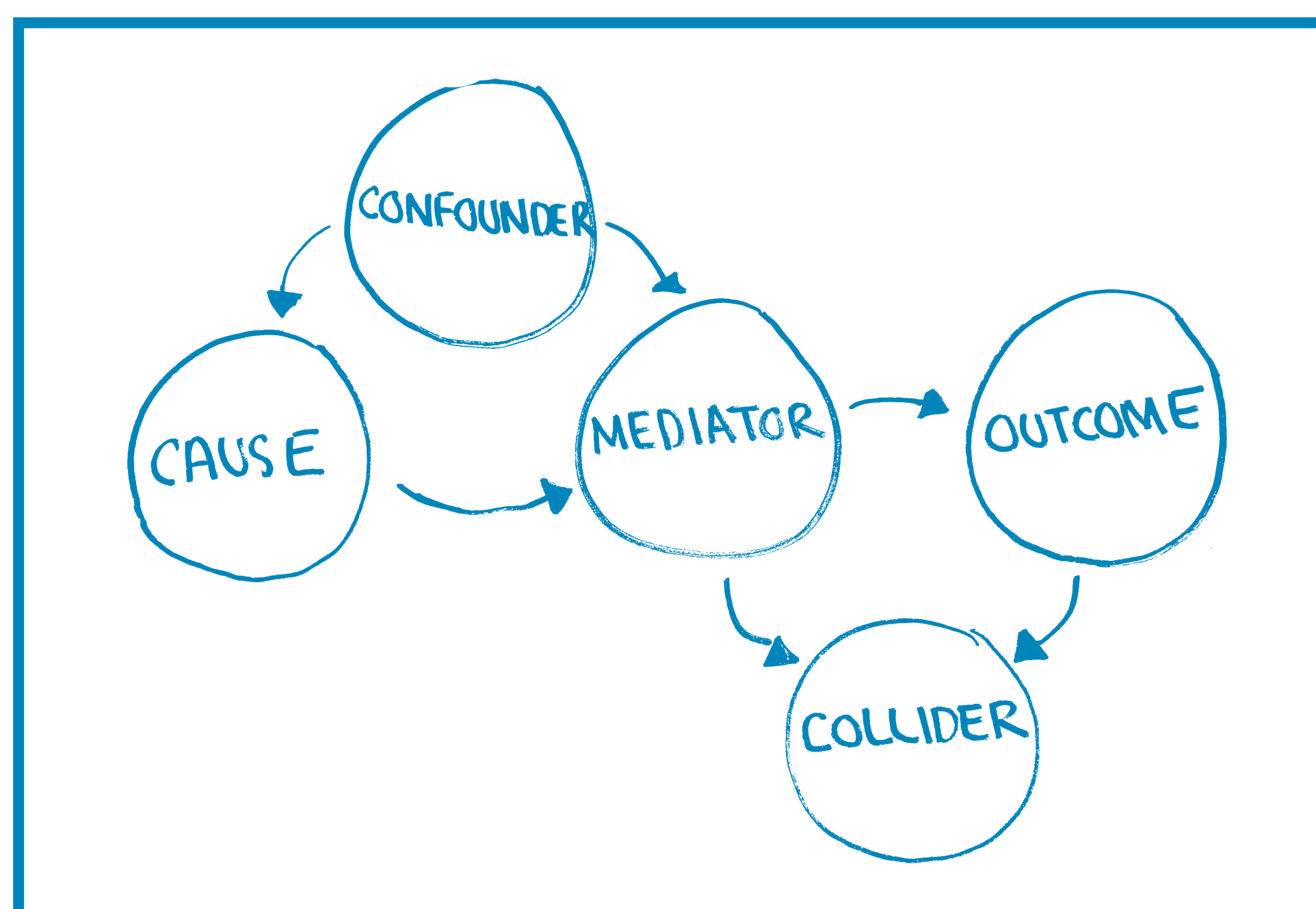
DAG-informed, expert-augmented Causal SHAP
recovers causally-coherent feature importance
that standard explainable ML misses.



The Truth



What SHAP Sees



THE CAUSAL SHAP WORKFLOW

- 1 Discover**
Run a causal-discovery algorithm on observed data to obtain a candidate DAG skeleton.
- 2 Resolve**
An expert orients the ambiguous edges using domain knowledge.
- 3 Weight**
Estimate the strength of each directed edge.
- 4 Compute**
Plug the resolved DAG into Causal SHAP and compute attributions.
- 5 Sensitivity**
Re-run Steps 2–4 under alternative plausible DAGs and compare.

Why standard SHAP leaks credit:
Every ordering gets an equal vote

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \left[f\left(S \cup \{i\}\right) - f\left(S\right) \right]$$

The sum runs over every permutation, including orderings where a mediator enters before its upstream cause. Averaging bakes the leakage into the attribution by design.

EXPERIMENTS

We ran two case studies.

We compared the Causal SHAP workflow against standard SHAP in two settings: a synthetic benchmark where the data-generating process was fully known, and a real-world ICU dataset for hospital-acquired pressure injury, where domain experts specified a causal structure that distinguishes severity drivers from downstream interventions like vasopressors.

➔ Synthetic (simcausal)

12-variable DAG, 28 edges, Treatment as largest driver. Kendall's $\tau = 0.42$ between Standard SHAP and Causal SHAP. 9 of 11 features re-ranked. The two methods agreed on more than half of rank-pairs but rearranged the ranking meaningfully.

➔ MIMIC-IV ICU

22,717 patients, 33 variables, expert DAG with 40 edges. Standard SHAP ranked vasopressors near the top. Causal SHAP demoted them and surfaced upstream severity drivers (lactate, mean arterial pressure, comorbidity burden) that better reflect the underlying causal structure.

TAKEAWAY

SHAP isn't lying. It's answering a narrower question than it sounds like.

SHAP decomposes a fitted model's predictions. That is not the same as ranking features by causal contribution, and not the same as ranking them by real-world predictive importance. The refinement process, expert and algorithm correcting each other, is what makes the explanation defensible.

Learn more about Causal SHAP and our case studies ➔

